

Rishita Jain

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EXPERIENCE

CSIR-IGIB

Dec. 2025 – May. 2026

Intern

Delhi, India

- Supported transcriptomics research by **optimizing Bash scripts** for automated processing of **multi-sample RNA-seq datasets**.
- Used **SnakeAltPromoter** to identify alternative promoters with potential **biomarker** relevance in **hepatocellular carcinoma (HCC)**.
- Contributed to identifying **4 genes** associated with HCC through **structured RNA-seq analysis**.
- Assisted with research documentation and interpretation of findings in a genomics-focused laboratory environment.

Omnison Pharma Pvt. Ltd.

May. 2024 – June. 2024

Intern

Ghaziabad, Intern

- Performed **analytical testing** of raw material samples for pharmaceutical formulation and manufacturing processes.
- Operated **UV-Vis spectrophotometry** and **HPLC** systems to assess purity and support quality assurance workflows.
- Processed an average of **3 samples per day** while maintaining accurate records and adhering to lab procedures.
- Gained exposure to regulated healthcare product workflows, documentation practices, and process reliability in an industry setting.

PROJECTS

Computational screening of plant-based peptides against carbapenemases

- Screened **PlantPepDB**, **DRAMP** and **CAMP4** to identify **negatively charged** antibacterial plant peptides targeting **Gram-negative bacteria**.
- Utilized various bioinformatics tools like **BLAST**, **protparam**, **AMP prediction**, **PyMol** for further shortlisting peptides on the basis of their **molecular weight**, **instability index**, **aliphatic index**, **GRAVY**, **half-life**, **antibacterial score**.
- Designed **mutant peptides** using amino acid walking method and **docked** them against the target OXAs using **Cluspro**.

Isolation and characterization of actinomycetes

- Isolated and identified ***Nocardia prasina*** through laboratory culture and characterization methods.
- Performed **biochemical testing** for **species characterization** and **functional assessment**.
- Conducted **antimicrobial screening** and observed maximum resistance against ***Klebsiella pneumoniae*** and ***Salmonella typhi***.

EDUCATION

Vellore Institute of Technology

Sept. 2022 – Present

Bachelor of Technology in Biotechnology

Vellore, India

- **CGPA**: 9.13

Delhi Public School

2021

Senior Secondary

Noida, India

- **Percentage**: 97.6

SKILLS

Languages : Python, JavaScript, R, bash

Tools : MS Excel, MS Powerpoint, MS Word, Figma, PowerBI

Bioinformatics : BLAST, Clustal Omega, MEGA 7, PyMol, Cluspro, proActiv, DEXSeq, Salmon, SnakeAltPromoter, STAR

Lab Techniques : UV-Vis Spectrophotometry, Gel Electrophoresis, HPLC, Titration, Microscopy, Antimicrobial screening, Agar plating

Soft Skills : Written and verbal communication, Teamwork, Self-learning, Critical thinking, Adaptability